INTEGRAL ATTACHMENTS AND WALL CORRESPONDENCES FOR K3 BUNDLE MODULI

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Abstract. We calculate the integral cohomology of the complements of the fibres over the completely split bundle in a compact six-chamber family of rank-three bundle moduli on a K3 surface, together with the boundary restrictions and the actions of the adjacent wall correspondences. Four crossings are elementary Mukai flops, with their corrected correspondences, and two are isomorphisms over divisorial wall targets. We identify the latter actions as reflections by proving the opposite signs of the exceptional divisor in the two Mukai markings. An explicit transport to the Hilbert square gives integral coordinates for the exceptional surfaces and all determinant-normalized universal indices. In these coordinates the complement ring is the quotient by the transported classes of the two planes in one chamber, and the degree-four boundary restriction is an explicit functional with values in the cyclic group of order eight. The formulas cover all six exceptional configurations and determine their cohomological wall compositions. Pairing the resulting cycles with the harmonic Kähler class identifies their periods and volumes in the established metric degeneration of this family.

1. Introduction

Wall crossing for bundles on a K3 surface relates compact holomorphic symplectic manifolds through contractions of exceptional subvarieties. The local quotient describes those subvarieties and their intersections. To calculate their attachments to the compact moduli space one must also identify their integral classes in its global cohomology ring. The same distinction arises for wall maps: a local change of exceptional configuration does not specify the action of a compact correspondence on universal classes.

We carry out these identifications for a rank-three family with three central line factors and cross-extension dimensions $1, 2, 1$. Its six nearby chambers have two types of deepest exceptional fibre: two planes meeting transversely at a point, or a plane and the first Hirzebruch surface meeting along its negative section. The compact family, its primitive exceptional cycles, and its harmonic metric degeneration were constructed in [Wue26a]. That calculation determined a middle-degree attachment with the marked complement ring as an input. The general descent statement in [Wue26b] likewise retains the geometric overlaps. Here we calculate the complement ring and all its ordinary integral cohomological attachment maps, and identify the correspondences induced by the actual compact slope walls.

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1.1. **The family and the main result.** Fix a marked projective K3 surface X with Néron–Severi lattice generated by H, D_1, D_2, where H is ample and the intersection matrix is

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & -6 & -4 \\ 0 & -4 & -8 \end{pmatrix}. \tag{1}$$

The marking of $H^2(X,\mathbb{Z}) = \Lambda = 3U \oplus 2E_8(-1)$ is chosen so that

$$H = e_1 + f_1, \qquad D_1 = e_2 - 3f_2, \qquad D_2 = -4f_2 + e_3 - 4f_3, \tag{2}$$

where $e_i \cdot f_i = 1$ in the three hyperbolic planes. Put

$$\begin{aligned} &a_0 = (1,0,1), \quad a_1 = (1,D_1,-2), \quad a_2 = (1,D_2,-3), \\ &v = a_0 + a_1 + a_2 = (3, D_1 + D_2, -4), \qquad w = a_0 + a_2. \end{aligned} \tag{3}$$

Vectors are written in the Mukai convention $v(F) = \mathrm{ch}(F)\sqrt{\mathrm{td}_X}$, with pairing $(r,c,s)\cdot(r',c',s') = c\cdot c' - rs' - r's$. Let M_σ be the compact moduli space of slope-stable bundles of vector v in a sufficiently small generic chamber of $H + sD_1 + tD_2$. Its central classifying map is denoted $\pi_\sigma : M_\sigma \to Z$. The geometric inputs defining this map are restated in Section 2. Let $o \in Z$ be the class of $\mathcal{O}_X \oplus \mathcal{O}_X(D_1) \oplus \mathcal{O}_X(D_2)$, and set

$$E_\sigma = (\pi_\sigma^{-1}(o))_{\mathrm{red}}, \qquad V_\sigma = M_\sigma \setminus E_\sigma.$$

The three wall coordinates are

$$\zeta(s,t) = \frac{2\pi}{3}\begin{pmatrix} -10 & -12 \\ 8 & 0 \\ 2 & 12 \end{pmatrix}\begin{pmatrix} s \\ t \end{pmatrix}, \qquad \zeta_0 + \zeta_1 + \zeta_2 = 0. \tag{4}$$

Chamber A has signs $(+,-,+)$.

For the integral answer let $Y = X^{[2]}$ and $R = H^*(Y,\mathbb{Z})$, with its complex orientation. Write $\mu : \Lambda \to R^2$ for the usual Hilbert-square map, δ for half the effective Hilbert–Chow divisor, and $A \in R^4$ for the class of subschemes containing a fixed point. Section 4 gives a finite integral basis and all products in R. Define

$$\begin{aligned} u_0 &= 7H - 2D_1 - 2D_2, & u_2 &= -5H + 2D_1 + D_2, \\ p_0 &= A + 3\delta^2 + \tfrac{1}{2}\big(\mu(u_0)^2 + 5\delta\mu(u_0)\big), \\ p_2 &= A + \delta^2 + \tfrac{1}{2}\big(\mu(u_2)^2 - 3\delta\mu(u_2)\big). \end{aligned} \tag{5}$$

The classes p_0, p_2 are integral, although these expressions use halves. The universal families will be normalized by the rank-one index of $y = (0,-f_2,0)$; the definition is given in (16).

Theorem 1.1. *For the marked compact family above the following statements hold.*

1. *The four adjacent crossings on $\zeta_0 = 0$ and $\zeta_2 = 0$ are elementary Mukai flops. The two crossings on $\zeta_1 = 0$ are isomorphisms over the corresponding partial wall target. In the Mukai markings their integral ring actions are respectively the identity and the reflection $s_{a_1}(z) = z + (z,a_1)a_1$. For a flop the ring action is induced by $[\Gamma] + [P \times P']$, where Γ is the graph closure and P, P' are the exceptional planes.*
2. *The explicit Mukai isometry*

$$g = e^{2H - D_1 - D_2} s_{(1,H,2)}, \qquad g(v) = (1,0,-1),$$

specifies oriented integral ring identifications $\Phi_\sigma : H^*(M_\sigma,\mathbb{Z}) \to R$. In these identifications

$$H^*(V_\sigma,\mathbb{Z}) = R/(p_0,p_2). \tag{6}$$

The groups are free of ranks $1, 23, 274, 21$ *in degrees* $0, 2, 4, 6$ *and zero otherwise. All determinant-normalized universal-index Chern classes are given by the topological formula in Proposition 4.1 and reduction modulo* (p_0, p_2).

(3) *Let* W_σ *be a regular neighborhood of* E_σ *and give* $L_\sigma = \partial W_\sigma$ *its outward orientation. In chamber* A*, the degree-four restriction to* $H^4(L_A, \mathbb{Z}) = \mathbb{Z}/8$ *is*

$$[z] \longmapsto \int_Y z(p_0 - 3p_2) \pmod 8. \tag{7}$$

Sections 5–6 give the neighborhood and boundary rings, all restriction, relative and connecting maps in each chamber, and the wall actions on their integral generators. For the fixed normalizer y*, the divisorial action on indices includes the inverse determinant line of the* $y - a_1$ *index. The six-chamber composition is the identity on integral cohomology.*

The first assertion supplies the geometric maps used in the other two. At an isolated wall the exhaustive factor list leaves two rigid factors with cross-extension dimension three. Their whole compact fibre is the plane in the incidence resolution. At the divisorial wall, the second factor varies over an entire smooth K3 surface, including the curve lying above o. Both resolutions are the intrinsic blow-up of the reduced singular surface in the partial wall target. A determinant section has a simple zero on the exceptional divisor; its opposite signs in the two Mukai markings force the reflection.

To calculate the integral maps we use the diagonal blow-up of $X \times X$ and classical integral Hilbert-square cohomology. Markman's universal correspondence for the displayed g identifies this calculation with the chamber spaces. The exceptional classes then have the coordinates (5). Primitive restriction witnesses identify the kernel of restriction to V_σ as their ideal. Poincaré–Lefschetz duality computes the boundary restrictions and fixes their connecting signs. This order of construction uses global K3 geometry before excision; local singularity types alone are not inputs sufficient to recover multiplication.

1.2. **Relation to earlier results.** The finite exceptional configurations already occur in Bielawski–Dancer [BD00, Example 6.10]. Integral cohomology of the corresponding unimodular hypertoric quotients is covered by Hausel–Sturmfels [HS02, Theorem 1.1]; their core intersection forms are treated by Hausel–Swartz [HS06, Theorem 2]. Compact configurations of planes and ruled surfaces under successive Mukai flops also occur in the K3 and EPW construction of Donten-Bury et al. [DGKKW17, §3.1 and Remark 6.17]. Thus neither the two local fibre types nor the existence of interacting compact exceptional surfaces is the assertion of this paper.

The integral basis used below is due to Qin–Wang [QW05, Theorem 5.4 and Remark 5.6], in the geometric form of Boissière–Nieper-Wißkirchen–Sarti [BNS13, Remark 5.7]. Markman's integral transport and universal-class compatibility [Mar08] provide the comparison with Y. The corrected flop correspondence is classical [Rie14, LLW10]; identity and reflection actions at K3 walls have their general Bridgeland counterparts in Bayer–Macrì [BM14, Lemma 10.1]. Our contribution is the compact slope-wall identification together with the full integral calculation in its specified marking, including the complement, the six attachments and the universal normalization. The quotient-ideal argument and the cohomological six-cycle relation are consequences of these identifications.

Finally Section 7 pairs the calculated classes with the harmonic Kähler class. This gives the signed line periods and ruled surface volumes and identifies the cycles in the metric theorems of [Wue26a, Wue26b]. The analytic estimates are imported with their hypotheses.

We make no claim about Chow-level path relations, monodromy of a complex smoothing, or reconstruction of an underlying manifold or metric tensor from the cohomology ring.

2. Compact moduli and their markings

We recall precisely the geometric results about the family that will be used. Their proofs are in [Wue26a, §§4–7]. All moduli spaces in this paper refer to this fixed complex K3 surface, rather than to an unspecified surface with the same intersection matrix.

2.1. **Central factors and compact contractions.** The embedding (2) is primitive, and the lattice (1) contains no roots. Indeed a root would give $x^2-3y^2-4yz-4z^2=-1$, which is impossible modulo four, whether y is even or odd. The K3 realization and the choice of ample component in [Wue26a, §4.1] therefore give the stated X, H. Let $L_0=\mathcal{O}_X$, $L_1=\mathcal{O}_X(D_1)$ and $L_2=\mathcal{O}_X(D_2)$. Then

$$((a_i,a_j))_{i,j=0}^2=\begin{pmatrix}-2&1&2\\1&-2&1\\2&1&-2\end{pmatrix},\qquad v^2=2,\quad w^2=0. \tag{8}$$

At H the cross Hom and second Ext groups of these distinct lines vanish. Riemann–Roch gives cross-extension dimensions $1,2,1$.

The complete central factor and charge calculation is [Wue26a, Lemma 4.1]. For vector v its polystable objects are the stable bundles, $L_1\oplus F$ with F stable of vector w, and $L_0\oplus L_1\oplus L_2$. For vector w the only strictly polystable object is $L_0\oplus L_2$. Every central semistable sheaf in both problems is locally free. The same calculation excludes all positive lost-charge ideal objects. Surface gauge compactness then excludes such objects throughout a sufficiently small neighborhood of H, including its walls [Wue26a, Lemma 3.2 and §4.2]. Thus the generic chamber spaces are full compact Gieseker moduli, consisting entirely of slope-stable bundles, and have deformation type $\mathrm{K3}^{[2]}$.

Here the assertion about positive charge is stronger than the absence of singular sheaves in a particular chamber. It gives smooth gauge subsequences when polarizations approach a wall. This is used below to identify whole exceptional fibres with finite quotient fibres.

The auxiliary generic w-moduli is a smooth compact K3 surface S. Its central map $S\to T$ contracts one (-2)-curve C to the class of $L_0\oplus L_2$. Write $S^\circ=S\setminus C$. The normal projective central target Z has the disjoint strata

$$Z=U\sqcup S^\circ\sqcup\{o\}, \tag{9}$$

where U is the connected dense stable open set. Its reduced singular surface is T. Each projective map $\pi_\sigma: M_\sigma\to Z$ is an isomorphism over U, has a projective-line fibre over every point of S°, and has the exceptional configuration described below over o. Projective and fixed-H gauge topologies agree.

We use the full convergent classifying charts of [Wue26a, §§2,4]. At o the effective compact group is $K=\mathrm{U}(1)^3/\mathrm{U}(1)$. With single arrows $z:1\to 0$, $w_1:1\to 2$ and a double arrow $u:0\to 2$, and reverse arrows z',w_1',φ, the equations are

$$\begin{gathered}zz'-\varphi(u)=0,\qquad w_1w_1'+\varphi(u)=0,\\ \mu_0=|z|^2-|z'|^2-|u|^2+|\varphi|^2,\qquad \mu_2=|w_1|^2-|w_1'|^2+|u|^2-|\varphi|^2,\\ \mu_1=-\mu_0-\mu_2,\qquad \mu=\zeta.\end{gathered} \tag{10}$$

The norms are the central harmonic norms after choosing orthonormal coordinates. These charts commute with the projective-to-affine quotient and the central graded-class map. They

cover whole fibres for small parameters, including moving fibres whose bases tend to o. Along S° the transverse quotient is the two-arrow A_1 model, together with its smooth surface tangent factor. The finite wall list is exactly $\zeta_i = 0$.

The whole-fibre statement has two parts. No-charge compactness and normalized limiting inclusion and quotient maps force a sequence over a given graded class to converge to that class. On the finite side, vanishing cycle invariants give an acyclic arrow support. Summing moments over its source and sink subsets bounds every arrow norm squared by a fixed multiple of $|\zeta|$. Thus its complete exceptional core lies in the same convergent chart. These are the coverage results proved in [Wue26a, §4.4]; a formal germ alone would not have this consequence.

2.2. **Cycle classes and integral witnesses.** Let $\theta_v : v^\perp \to H^2(M_\sigma, \mathbb{Z})$ be the Mukai homomorphism, in the convention

$$\theta_v(z) = \Big[p_{M*}\big(\mathrm{ch}(\mathcal{E})\sqrt{\mathrm{td}_X}\, z^\vee\big)\Big]_2 . \tag{11}$$

Here $p_M : X \times M_\sigma \to M_\sigma$, $\mathcal{E}$ is a universal class and $(r,c,s)^\vee = (r,-c,s)$. The map is independent of twisting $\mathcal{E}$ by a line bundle from M_σ. We identify curve classes with rational elements of $v^\perp$ through the Mukai pairing and θ_v. In chamber A the primitive line classes in the two planes are

$$c_0 = v/2 - a_0, \qquad c_2 = v/2 - a_2, \qquad f = c_0 + c_2 = a_1. \tag{12}$$

The last class is the proper reduced specialization of a moving $\mathbb{P}^1$ fibre. The planes meet transversely once. Their normal bundles are their cotangent bundles, and their classes satisfy

$$[P_i] = \tfrac{1}{2}\theta_v(c_i)^2 + \tfrac{1}{24}c_2(M_\sigma), \qquad \left(\int [P_i][P_j]\right)_{i,j=0,2} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}. \tag{13}$$

These are the geometric identifications of [Wue26a, §§5–6]; the plane formula uses the classical K3$^{[2]}$ intersection theory of [HT09].

We will need integral, rather than rational, independence. Define

$$x = (0, f_2 - f_3, 0), \qquad y_0 = (0, -3f_2, -1) \quad \text{in } v^\perp. \tag{14}$$

The universal extensions on the planes give, with their positive hyperplane classes h_0, h_2,

$$\begin{aligned} \theta_v(x)|_{P_0\cup P_2} &= (0, h_2), \quad \theta_v(y_0)|_{P_0\cup P_2} = (-h_0, -h_2), \\ \theta_v(x)^2|_{P_0\cup P_2} &= (0, h_2^2), \quad \theta_v(y_0)^2|_{P_0\cup P_2} = (h_0^2, h_2^2). \end{aligned} \tag{15}$$

Both restriction matrices are unimodular. They prove that the pairs of plane and line classes embed primitively, and that the two evaluation maps onto $\mathbb{Z}^2$ are surjective.

2.3. **Universal indices and twisting.** The v-moduli spaces admit ordinary universal families, unique up to a base line bundle, since $\chi(v) = -1$. The auxiliary w-moduli has a nonzero algebraic Brauer obstruction of order two. We retain local twisted universal families there. This obstruction is not removed by constancy of the fibre cohomology of its projective extension family.

Identify $K^0_{\mathrm{top}}(X)$ with the Mukai lattice by its Mukai vector. This convention lets a symbol such as $y = (0, -f_2, 0)$ denote both an integral vector and the corresponding topological class; θ_v takes the vector, whereas a family index takes the class. For any $z \in K^0_{\mathrm{top}}(X)$ put

$$\begin{aligned} I_z(\mathcal{E}) &= p_{M!}(p_X^* z^\vee \cdot \mathcal{E}), \qquad y = (0, -f_2, 0), \\ \mathcal{E}^y &= \mathcal{E} \otimes p_M^*(\det I_y(\mathcal{E}))^{-1}, \qquad I_z^y = I_z(\mathcal{E}^y). \end{aligned} \tag{16}$$

The exclamation mark denotes the complex-oriented topological K-theory pushforward. Since $\chi(v,y) = -(v,y) = 1$, the y-index has rank one. The normalization cancels any base line twist of $\mathcal{E}$, and $\det I_y^y$ is trivial. Its restrictions to the planes have the particularly useful form

$$c(I_z^y)|_{P_i} = (1+h_i)^{m_i(z)}, \qquad m_i(z) = \chi(L_i, z) = -(a_i, z), \quad i = 0, 2. \tag{17}$$

This is the normalized extension calculation in [Wue26a, §6.4]; it also holds for virtual and nonalgebraic topological classes. Section 4 computes all these indices globally from the universal subscheme on $X^{[2]}$.

3. The compact wall correspondences

The central factor classification determines the partial wall targets as well as the central target. We first identify these compact maps, then compute their actions. This is where the global moduli problem enters the wall calculation.

3.1. **Partial targets and whole fibres.** Choose a small rational nonzero parameter $\eta_* = (s_*, t_*)$ on one wall $\zeta_i = 0$. Choose generic polarizations in the adjacent sectors, close enough that there is no intervening wall. The finite wall theorem and the exhaustive nearby list imply that every bundle in either chamber is η_*-semistable. Normalize the common image of their projective slope classifying maps. As in the central construction, connected fibres and normal-function descent give a normal projective partial target Z_i and morphisms

$$M_+ \longrightarrow Z_i \longleftarrow M_-. \tag{18}$$

Their fibres are the η_*-graded classes. This uses the same slope classifying construction of Greb–Toma [GT17] and the whole-fibre descent in [Wue26a, §4.5].

There are no additional equal-slope decompositions on this partial wall. Indeed the H-Jordan–Hölder classes of a proper equal-slope factor must be a subcollection of a_0, a_1, a_2, or a varying w-factor. Its equality equation is the corresponding subset sum of the ζ_i. At a nonzero point just one of these three equations vanishes. The bounded wall theorem excludes further walls after shrinking the neighborhood of H. Local freeness and the exclusion of lost charge persist on the partial wall by the central factor and charge result recalled above.

Proposition 3.1. *If $i = 0$ or 2, the two morphisms in (18) are the two resolutions of one isolated elementary Mukai contraction; their birational map is an elementary Mukai flop. If $i = 1$, the reduced singular locus of Z_1 is a smooth compact K3 surface, and both morphisms are its intrinsic blow-up. In particular they define a unique isomorphism $M_+ \simeq M_-$ over Z_1.*

Proof. For $i = 0, 2$, the only strictly polystable class is $L_i \oplus G$, where G has vector $v - a_i$. The other two line factors have unequal η_*-slopes. Their one-dimensional cross Ext gives the unique nontrivial extension in the stable direction. It is stable: a destabilizing subsheaf would, by the exhaustive nearby wall list, give the wrong ordered line subfactor. Its vector is spherical. Uniqueness also follows from stability and Riemann–Roch: two nonisomorphic stable sheaves of the same spherical vector have zero Hom in both directions and hence nonpositive Euler characteristic, whereas that characteristic is two.

The factors L_i, G have equal slope and are stable of different ranks. Thus cross Hom and second Ext vanish, and

$$\dim \operatorname{Ext}^1(L_i, G) = (a_i, v - a_i) = 3.$$

They are rigid, so the full normal quotient is the doubled three-arrow quotient without a tangent factor. Its two resolutions are the incidence resolutions of the cone of rank-one square-zero endomorphisms of $\mathbb{C}^3$.

These are whole compact fibres. To see why the local assertion suffices here, take a contrary sequence of points in a wall fibre outside its fixed classifying chart as the transverse parameter tends to zero. No-charge compactness gives a smooth gauge limit. Normalize the extension inclusion or quotient in L^2 and pass to its nonzero limiting map. Equal-slope polystability splits off the corresponding stable factor, and the determinant and factor list identify the other. The limit is $L_i \oplus G$, contradicting exclusion from the chart. The finite zero-section core has radius of order the square root of the transverse moment parameter, so the whole finite plane is in this chart as well. The two classifying maps are isomorphisms off that fibre. The common blow-up is locally the total space of $\mathcal{O}(-1,-1)$ over the incidence flag variety in $\mathbb{P}^2 \times (\mathbb{P}^2)^*$. Its flag variety is the exceptional divisor of the four-dimensional blow-up. Gluing to the common regular open set proves the compact elementary-flop assertion.

For $i = 1$, write $A_1 = L_1$. The polystable objects are $A_1 \oplus F$ with F stable of vector w at η_*. Since $\zeta_0 = -\zeta_2 \neq 0$, this is a generic polarization for the auxiliary problem. Its full moduli is the smooth compact K3 surface S, including the curve over the central split class. Equal slopes and different ranks imply cross Hom vanishing at every point of S. Serre duality therefore gives constant cross-extension dimension $(a_1, w) = 2$.

The full classifying model along this locus is the product of a smooth surface germ with $\mathbb{C}^2/\{\pm 1\}$. The effective scalar stabilizer acts trivially on the tangent directions to S. The same compactness and normalized-map argument covers the entire $\mathbb{P}^1$ fibre over each point of S; compactness of S gives a finite cover of these models. The reduced singular locus is S, and there is no deeper singularity on this nonzero wall.

The minimal resolution of the A_1 surface is the blow-up of its reduced singular point. Blow-up commutes with the smooth local product, so both maps are locally $\mathrm{Bl}_S Z_1 \to Z_1$. Their identifications agree on the dense regular set and hence on overlaps. They glue to a unique global identification with the intrinsic blow-up. Proper GAGA makes this identification algebraic. Local universal w-families may be twisted, but all these constructions use their coarse classifying spaces and descend; no splitting of the algebraic Brauer class is required. The resulting isomorphism also commutes with the maps to Z, since its two composites agree on the dense stable open and Z is separated. □

3.2. **The divisor sign and the reflection.** Use $+$ for the side $\zeta_1 < 0$ and $-$ for $\zeta_1 > 0$ in this subsection. Let $D_\pm$ be the exceptional divisor of the maps to Z_1, and let $\theta_\pm$ denote the Mukai markings.

Lemma 3.2. *The two effective divisor classes are*

$$[D_+] = \theta_+(a_1), \qquad [D_-] = -\theta_-(a_1).$$

Both determinant sections have multiplicity one.

Proof. The first formula is the determinant calculation of [Wue26a, §7.2], applied over the full auxiliary S at this partial wall. We give the opposite-side argument explicitly. On the minus side $\mu(A_1) < \mu(E)$, so $\mathrm{Hom}(E, A_1) = 0$. The locus $\mathrm{Hom}(A_1, E) \neq 0$ is exactly D_-. For a nonzero map, let B be the saturated rank-one hull of its image. The quotient E/B is torsion-free, so the depth lemma on the smooth surface makes B reflexive, hence a line bundle. The effective zero divisor of $A_1 \to B$ has degree zero by H-semistability of E; ampleness of H forces it to vanish. Thus $B = A_1$. The torsion-free quotient F is H-semistable of vector w, and [Wue26a, Lemma 4.1] makes it locally free. It is semistable at the partial wall because E is and A_1 has the same wall slope; the auxiliary polarization is generic, so F is stable there. Conversely every nontrivial extension on this side has an inclusion of A_1.

The perfect complex $Rp_*R\operatorname{Hom}(A_1,\mathcal{E})$ has amplitude $[0,1]$ and virtual rank zero, since $(a_1,v)=0$. Its determinant section is a section of the inverse determinant line. The first Chern class of that line is $-\theta_-(a_1)$. At an extension $0\to A_1\to E\to F\to 0$ the kernel has dimension one: $\operatorname{Hom}(A_1,E)=\mathbb{C}$, and hence $\dim\operatorname{Ext}^1(A_1,E)=1$. The derivative of the determinant is the obstruction to preserving the inclusion. The map

$$\operatorname{Ext}^1(E,E)\longrightarrow\operatorname{Ext}^1(A_1,E) \tag{19}$$

is surjective. In the long exact sequence its cokernel maps into $\operatorname{Ext}^2(F,E)\to\operatorname{Ext}^2(E,E)$, which is an isomorphism: the Serre-dual map $\operatorname{Hom}(E,E)\to\operatorname{Hom}(E,F)$ is generated by the quotient and is an isomorphism. Thus a local two-term presentation of the index has corank one with a nonzero normal derivative of its determinant. The zero is simple. This proves the second formula with multiplicity one; the dual inclusion/quotient argument gives the same multiplicity on the plus side. □

Proposition 3.3. *In the Mukai markings the isomorphism across $\zeta_1=0$ acts on degree two by s_{a_1} and on the integral ring by Markman's $\gamma_{s_{a_1}}$. Across $\zeta_0=0$ or $\zeta_2=0$, the corrected flop correspondence acts by γ_{id}.*

Proof. Let $f:M_+\to M_-$ be the divisorial isomorphism and set $\psi=\theta_-^{-1}f_*\theta_+$. The universal families agree up to a base line on the common stable open. Its complement is the connected smooth prime divisor D_-, so the kernel of restriction in degree two is $\mathbb{Z}[D_-]$, by the Thom sequence. Lemma 3.2 implies $\psi(a_1)=-a_1$, while $\psi(x)-x$ is an integral multiple of a_1 for every $x\in v^\perp$. Write it as ka_1. Preservation of the Beauville–Bogomolov form gives

$$(x,a_1)=(\psi x,\psi a_1)=-(x,a_1)+2k,$$

so $k=(x,a_1)$, as asserted.

For an isolated plane wall let Γ be the graph closure. Rieß [Rie14, Theorem 3.2, Lemma 3.5 and §6] gives the integral multiplicative correspondence

$$\mathcal{Z}=[\Gamma]+[P\times P'],\qquad \mathcal{Z}_*z=\Gamma_*z+\left(\int_P z\right)[P']\quad(z\in H^4). \tag{20}$$

Its transpose is the inverse. On the complement of the plane the universal families agree up to a base line. Deleting complex codimension two does not change degree-two cohomology, so $\mathcal{Z}_*$ is the identity in the Mukai markings. The rational cohomology of a $\mathrm{K3}^{[2]}$ fourfold is generated by degree two, as is also immediate from the formulas in Section 4. Thus both ring maps agree rationally with the indicated Markman maps. Their integral cohomology is torsion-free, so the agreement is integral. This reasoning uses the corrected correspondence, not the graph alone. □

3.3. **The six effective configurations.** The cyclic order of the chambers and their wall maps is shown in Figure 1. The complete effective cycle table is Table 1. The notation p_i here means the class obtained from the plane formula (13); Section 4 identifies it with (5) in the common ring R.

Here is the geometric calculation of the table. If P_2 is flopped, its graph image is $-2[P_2']$ and the exceptional product contributes $3[P_2']$; see [LLW10, Example 6.6] or [Rie14, §6]. Thus the corrected image is $[P_2']$. The other plane meets P_2 transversely once. Its graph image is its blow-up at that point, and the product term adds $[P_2']$. Consequently

$$p_2\longmapsto[P_2'],\qquad p_0\longmapsto[\Sigma]+[P_2'],\qquad \Sigma\simeq\mathbb{F}_1. \tag{21}$$

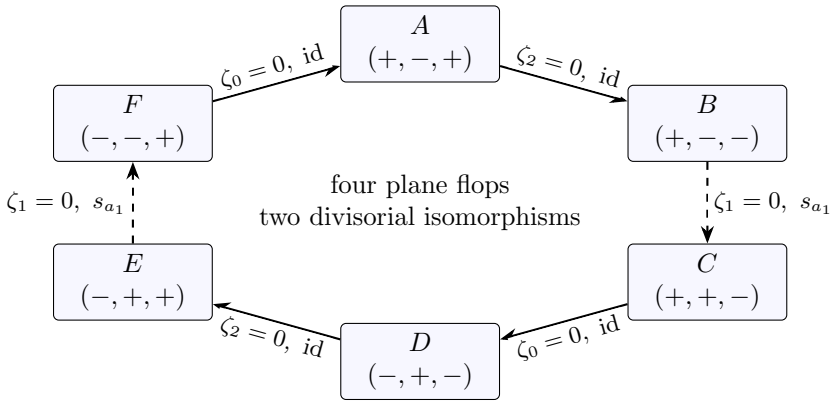


FIGURE 1. The chamber adjacency cycle. Solid edges are elementary Mukai flops with their corrected cohomology correspondences; dashed edges are the divisorial isomorphisms. The edge labels give the vanishing coordinate and the Mukai action.

TABLE 1. Effective deepest surfaces and curves. In a ruled row the last three entries are the plane line, the common negative section, and the ruling. Here $f = c_0 + c_2$.

Chamber	Signs $(\zeta_0, \zeta_1, \zeta_2)$	Surfaces	Curves
A	$(+,-,+)$	p_0, p_2	c_0, c_2 (two planes)
B	$(+,-,-)$	$p_2, p_0 - p_2$	$-c_2;\ -c_2;\ f$
C	$(+,+,-)$	$p_0, p_2 - p_0$	$c_0;\ c_0;\ -f$
D	$(-,+,-)$	p_0, p_2	$-c_0, -c_2$ (two planes)
E	$(-,+,+)$	$p_2, p_0 - p_2$	$c_2;\ c_2;\ -f$
F	$(-,-,+)$	$p_0, p_2 - p_0$	$-c_0;\ -c_0;\ f$

The common plane line and negative section represent $-c_2$. A line in P_0 missing the blown-up point keeps its class; on $\mathbb{F}_1$ it is the sum of negative section and ruling. The ruling therefore has class $f = c_0 + c_2$. This gives row B and the intersection matrix

$$\begin{pmatrix} p_2^2 & p_2(p_0 - p_2) \\ p_2(p_0 - p_2) & (p_0 - p_2)^2 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ -2 & 4 \end{pmatrix}. \tag{22}$$

The reflection sends c_0 to $-c_2$, c_2 to $-c_0$, and f to $-f$; it exchanges p_0, p_2 and preserves the tangent Chern class. This proves row C. The next flop gives the two-plane row D, with both line signs reversed. Repeating gives E, F and returns to A.

Corollary 3.4. *The chosen cohomological correspondences along any two paths in the six-chamber adjacency graph with the same endpoints agree. The complements V_σ are isomorphic over $Z \setminus \{o\}$, and the wall actions induce their cohomology identifications.*

Proof. The six forward actions in the Mukai marking are $\mathrm{id}, s_{a_1}, \mathrm{id}, \mathrm{id}, s_{a_1}, \mathrm{id}$. Their product is the identity. Each reverse edge is the inverse correspondence. Cancellation of reverse edges and insertion of complete six-cycles generate the relations between paths in this graph. The ring assertion follows. For the complements, the central map is an isomorphism over U and has the unique relative minimal A_1 resolution along S°. The local identifications agree on the dense regular locus and glue over $Z \setminus \{o\}$. □

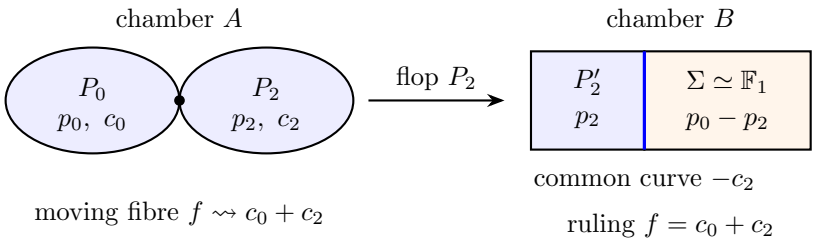


FIGURE 2. The two configurations and their classes across the flop from A to B. The right-hand common curve is a line in P_2' and the negative section of Σ. The moving fibre specializes in A to the reduced sum $c_0 + c_2$; in B this class is the ruling of Σ. The drawings indicate incidences and cycle classes, not metric shapes.

The corollary concerns these ordinary cohomological actions. It does not identify compositions of Chow correspondences or derived functors, and it is not a monodromy calculation for a smoothing.

4. Integral coordinates and universal classes

We now give the integral coordinates in which the compact wall and attachment calculations can be performed. The Hilbert-square basis and Markman transport are classical inputs. The purpose of this section is to fix their signs and the universal normalization for this moduli problem, including arbitrary topological index classes.

4.1. **The diagonal construction.** Let $\alpha_1, \dots, \alpha_{22}$ be an integral basis of Λ, write $q_{ij} = \alpha_i \cdot \alpha_j$, and let (q^{ij}) be its inverse matrix. Let x_X be the oriented point class of X. Consider

$$b : B = \mathrm{Bl}_\Delta(X \times X) \longrightarrow X \times X, \qquad p : B \longrightarrow Y = X^{[2]}.$$

The map p is the double cover ramified along the exceptional divisor E_B. Write $j : E_B \to B$, $\pi : E_B \to X$ and $e = [E_B] = p^*\delta$. A tensor class below is pulled back by b, and $e\alpha$ means $j_*\pi^*\alpha$. The blow-up relations are

$$\begin{aligned} (e\alpha)(u \otimes v) &= e(\alpha u v), \qquad (e\alpha)(e\beta) = -[\Delta](\alpha\beta \otimes 1), \\ [\Delta] &= x_X \otimes 1 + 1 \otimes x_X + \sum_{i,j} q^{ij} \alpha_i \otimes \alpha_j. \end{aligned} \tag{23}$$

For the sign, $E_B|_{E_B} = -\xi$ and the excess formula gives $b^*[\Delta] = j_*\xi$, since $c_1(TX) = 0$. Hence $e^2 = -b^*[\Delta]$. In particular $e^4 = 24 x_X \otimes x_X$. Integration on Y is half integration of a pullback on B; $p^*[\mathrm{pt}] = 2x_X \otimes x_X$. Thus (23) is a multiplication rule using only the marked K3 pairing.

The pullback of $\mu(\alpha)$ is $\alpha \otimes 1 + 1 \otimes \alpha$. The classes $\mu_i = \mu(\alpha_i)$ and δ form an integral basis in degree two. We extend q to their Beauville–Bogomolov pairing by

$$q(\mu(a), \mu(b)) = q(a, b), \quad q(\mu(a), \delta) = 0, \quad q(\delta, \delta) = -2. \tag{24}$$

An integral degree-four basis consists of the classes whose pullbacks are

$$\begin{aligned} p^*A &= x_X \otimes 1 + 1 \otimes x_X, & p^*T_i &= 2e\alpha_i, \\ p^*C_{ij} &= \alpha_i \otimes \alpha_j + \alpha_j \otimes \alpha_i \quad (i < j), & p^*Q_i &= \alpha_i \otimes \alpha_i - e\alpha_i. \end{aligned} \tag{25}$$

Integrality is Qin–Wang's theorem, with its K3 extension [QW05, Theorem 5.4 and Remark 5.6], in the geometric Nakajima conventions of [BNS13, Remark 5.7]. In particular Q_i is the half difference of the two Nakajima terms, not half a symmetric square by itself. These statements apply to all of Λ, including its nonalgebraic classes. We use the basis statement in that reference; the product signs here follow directly from (23).

For completeness we print the entire middle pairing. Degree-eight products in the following display denote their integral against Y:

$$\begin{aligned} A^2 &= 1, & AT_i = AC_{ij} = AQ_i &= 0, \\ T_iT_j &= -2q_{ij}, & T_iQ_j &= q_{ij}, \quad T_iC_{jk} = 0, \\ Q_iQ_j &= (q_{ij}^2 - q_{ij})/2, & Q_iC_{jk} &= q_{ij}q_{ik}, \\ C_{ij}C_{kl} &= q_{ik}q_{jl} + q_{il}q_{jk}. \end{aligned} \tag{26}$$

Every entry follows by multiplication on B and division by two. The pairing is unimodular. Eliminate the T block $-2q$ over $\mathbb{Q}$. Its Schur complement is half the tensor-product pairing on the symmetric tensors in $\Lambda \otimes \Lambda$; its determinant is $2^{-22}(\det q)^{23}$. To verify this identity, diagonalize q over a field extension: the diagonal symmetric tensors have entries $d_i^2/2$, and the off-diagonal ones have entries d_id_j. The product is the stated expression, and the identity descends over $\mathbb{Q}$. The determinant of the whole pairing is therefore $\det(-2q)2^{-22}(\det q)^{23} = (\det q)^{24} = 1$. This also verifies index one, once integrality and the known free rank 276 are supplied.

Products of degree-two classes are

$$\begin{aligned} \mu_i\mu_j &= q_{ij}A + C_{ij} && (i<j), \\ \mu_i^2 &= q_{ii}A + 2Q_i + T_i, & \mu_i\delta &= T_i, \\ \delta^2 &= -A - \sum_i q^{ii}Q_i - \tfrac{1}{2}\sum_i q^{ii}T_i - \sum_{i<j} q^{ij}C_{ij}. \end{aligned} \tag{27}$$

The diagonal of q^{-1} is even because the dual K3 lattice is even. Thus the last expression is integral. If α^i is the q-dual basis, set

$$R_i = \mu(\alpha^i)A, \qquad R_\delta = -\delta A. \tag{28}$$

These classes are dual to μ_i, δ and form an integral degree-six basis. For $a \in R^2$ and $z \in R^4$, its product az has coefficient $\int_Y \mu_i az$ on R_i and $\int_Y \delta az$ on R_δ. These are given by (27) and (26). Together with the unit, the oriented point, the dual degree-two/ degree-six pairing, and zero odd groups, this specifies every product. The ranks are $1, 23, 276, 23, 1$. Torsion-freeness is the integral Hilbert-scheme theorem; see [Tot20, Theorem 3.1].

The formulas also exhibit rational generation by degree two:

$$A = \frac{1}{10}\delta^2 + \frac{1}{20}\sum_{i,j} q^{ij}\mu_i\mu_j, \tag{29}$$

after which (27) gives T_i, C_{ij}, Q_i, and (28) gives degree six. The integral ring generally requires more than the integral symmetric square.

We will also use the classical tangent-class identity

$$c_2(Y) = 24A - 3\delta^2 \tag{30}$$

of Shen–Vial [SV16, Lemma 14.4]. Their incidence-surface class is A here. Their rational Chow identity gives this integral cohomological identity by torsion-freeness. The preceding

products give, independently,

$$\int_Y \delta^4 = 12, \quad \int_Y c_2\mu(a)\mu(b) = 30q(a,b), \quad \int_Y c_2\delta^2 = -60, \quad \int_Y c_2^2 = 828.$$

4.2. **Transport from the rank-three moduli.** Set $a_H = (1, H, 2)$, $C_0 = D_1 + D_2 - 2H$ and

$$g = e^{-C_0} s_{a_H}, \qquad s_{a_H}(z) = z + (z, a_H)a_H. \tag{31}$$

Here $e^D(r, c, s) = (r, c + rD, s + cD + rD^2/2)$. The input lattice gives

$$a_H^2 = -2, \quad (v, a_H) = -2, \quad C_0^2 = -14, \quad s_{a_H}(v) = (1, C_0, -8), \quad g(v) = v' = (1, 0, -1).$$

This is the integral Mukai isometry induced by the spherical twist about $\mathcal{O}_X(H)$ followed by tensoring with $\mathcal{O}_X(-C_0)$. It need not send each chamber sheaf to a sheaf.

The polarizations may be chosen in each sector with degree coprime to three, as in [Wue26a, §4.5]; the full compact Gieseker moduli therefore satisfies Markman's suitability condition [Mar08, Definition 3.4]. His universal Chern correspondence [Mar08, Theorem 1.2 and equation (3)] gives

$$\Phi_\sigma = \gamma_{g,v} : H^*(M_\sigma, \mathbb{Z}) \xrightarrow{\sim} R. \tag{32}$$

The source and target are torsion-free: for the source use the K3$^{[2]}$ deformation type established in Section 2, and for the Hilbert square use [Tot20, Theorem 3.1]. Thus the modulo-torsion qualification loses nothing here. The correspondence preserves tangent Chern classes and orientation; for the latter, preservation of $c_4 = 324[\mathrm{pt}]$ suffices. Compatibility with the Mukai homomorphisms is [Mar08, Theorem 3.9 and diagram (41)]:

$$\Phi_\sigma \theta_v(z) = \theta_{v'}(gz), \qquad \theta_{v'}(r, c, r) = \mu(c) + r\delta. \tag{33}$$

The sign in the second formula follows from the universal ideal sequence: $c_1(\mathcal{O}_X^{[2]}) = -\delta$, while the index of $(r, -c, r)$ is its Euler characteristic minus its tautological class. This determines Φ_σ without choosing an unspecified deformation path.

4.3. **Topological indices and determinant normalization.** Let $\mathcal{I}$ be the universal ideal class on $X \times Y$. For a topological class F let $F^{[2]}$ denote the index defined by the universal length-two subscheme. With g acting on topological classes through their Mukai vectors, put

$$J_z = \chi_X((gz)^\vee)\, 1 - ((gz)^\vee)^{[2]}, \qquad \mathcal{L} = \det J_y, \qquad \widehat{J}_z = J_z \otimes \mathcal{L}^{-1}. \tag{34}$$

Here $gy = (0, -f_2, 1)$ and $\operatorname{rank} J_y = 1$.

Proposition 4.1. *For every $z \in K^0_{\mathrm{top}}(X)$ and $k \geq 0$,*

$$\Phi_\sigma(c_k(I_z^y)) = c_k(\widehat{J}_z). \tag{35}$$

These classes are calculated in the integral basis above by

$$p^* \operatorname{ch}(F^{[2]}) = \operatorname{ch}(F)_1 e^{-e} + \operatorname{ch}(F)_2, \qquad F \in K^0_{\mathrm{top}}(X). \tag{36}$$

Subscripts denote the two pullbacks from X to B.

Proof. Markman's universal-class identity intertwines the universal Chern characters up to a topological base line bundle. Contract its K3 factor against gz using the Euler pairing. Since g is an Euler isometry, the index characters are intertwined up to that same line. Taking the determinant of the rank-one y-index multiplies it by exactly this line, so normalization cancels it. The resulting normalized characters agree over $\mathbb{Q}$, and hence so do the Chern classes. Both Chern classes are integral and $H^*(Y, \mathbb{Z})$ is torsion-free, proving (35) integrally.

We prove (36) for topological F, without requiring an algebraic representative of its first Chern class. Let $\mathcal{Z} \subset Y \times X$ be the universal subscheme and pull it back to $\mathcal{Z}_B \subset B \times X$. The incidence description is the diagonal blow-up [SV16, §10, diagram (54)]. If $q_1, q_2 : B \to X$ are the projections, $\mathcal{Z}_B$ is the scheme-theoretic union of their graphs, intersecting in the graph over E_B. This includes the nonreduced fibre: in a blow-up chart with difference coordinates (t, tw) the length-two algebra is $A[U]/(U(U-t))$. Evaluation at $0, t$ maps it injectively to $A \oplus A$, with image the pairs agreeing modulo t and cokernel $A/(t)$. The other blow-up chart gives the same description. Thus there is an exact sequence of kernels

$$0 \longrightarrow \mathcal{O}_{\mathcal{Z}_B} \longrightarrow \mathcal{O}_{\Gamma_1} \oplus \mathcal{O}_{\Gamma_2} \longrightarrow \mathcal{O}_{\Gamma_{E_B}} \longrightarrow 0. \tag{37}$$

All these kernels are perfect on the smooth product $B \times X$. Pass their equality to topological K-theory, tensor with the pullback of any F, and push forward to B. Base change for the complex-oriented projection and functoriality give

$$p^*F^{[2]} = q_1^*F + q_2^*F - j_!\pi^*F = q_1^*F \otimes \mathcal{O}_B(-E_B) + q_2^*F.$$

The last equality uses the divisor Koszul class $j_!1 = 1 - [\mathcal{O}_B(-E_B)]$ and the projection formula. Only the fixed kernel in (37) needed an algebraic representative. The equality therefore holds for arbitrary topological F. Applying the Chern character proves (36). □

For use as a finite formula, a Mukai vector (r, c, s) has ordinary character $r + c + (s-r)x_X$. Truncate the exponential in (36) after degree eight, form (34), and multiply its character by $\exp(-c_1(J_y))$. The Newton identities recover c_1 through c_4; higher classes vanish on Y. Formula (23) and the integral bases then give their coordinates. This covers virtual classes of every rank. The intermediate arithmetic is rational, but the results are Chern classes of integral topological indices.

4.4. **Exceptional coordinates.** Applying (31) to (12) gives

$$gc_0 = (5/2, u_0, 5/2), \qquad gc_2 = (-3/2, u_2, -3/2), \qquad q(u_0) = 10, \quad q(u_2) = 2. \tag{38}$$

Substituting (33) and (30) in the plane formula gives exactly (5). For any $a \in \Lambda$, $\mu(a)^2 - \delta\mu(a)$ is even by (27) and the evenness of q. This proves the asserted integrality of both plane coordinates. Multiplication gives $p_0^2 = p_2^2 = 3$ and $p_0 p_2 = 1$.

Write

$$\begin{aligned} &\lambda_0 = \mu(u_0) + \tfrac{5}{2}\delta, \qquad \lambda_2 = \mu(u_2) - \tfrac{3}{2}\delta, \qquad d_i(a) = q(\lambda_i, a), \\ &\ell_0 = \sum_j q(u_0, \alpha_j) R_j - 5R_\delta, \qquad \ell_2 = \sum_j q(u_2, \alpha_j) R_j + 3R_\delta. \end{aligned} \tag{39}$$

The functionals $d_i : R^2 \to \mathbb{Z}$ are integral, and ℓ_i is their Poincaré dual. They are precisely the line functionals and line classes of the two planes under Φ_A. The primitive witnesses (15) therefore apply to these explicit coordinates.

5. The complement and its integral attachments

All coordinates in this section refer to the ring R of Section 4. In particular the ambient multiplication and the exceptional classes have already been calculated. We use excision to determine the complement and its maps, with integral coefficients throughout.

5.1. **The exceptional ideal.** First work in chamber A, with $E = P_0 \cup P_2$ and $V = M_A \setminus E$. The planes meet at one point, so a regular neighborhood retracts to $\mathbb{P}^2 \vee \mathbb{P}^2$. Integral Poincaré–Lefschetz duality on the regular neighborhood, followed by excision [Hat02, Theorem 3.43], gives

$$H^k(M_A, V) \simeq H_{8-k}(E). \tag{40}$$

The only nonzero relative groups are in degrees $4, 6, 8$, with ranks $2, 2, 1$. Their maps to the absolute cohomology are the plane, line and point inclusions, respectively.

Proposition 5.1. *Restriction is a surjective ring homomorphism with kernel (p_0, p_2). Hence $H^*(V, \mathbb{Z}) = R/(p_0, p_2)$, with free ranks $1, 23, 274, 21$ in degrees $0, 2, 4, 6$ and no other groups.*

Proof. The witnesses (15) show that the relative plane and line maps in (40) are primitive injections. The point map is an isomorphism. The long exact sequence of the pair therefore gives surjectivity of restriction, vanishing of odd groups, and the stated free quotients. It remains to identify its kernel as an ideal, rather than merely as three additive subgroups.

The projection formula on the two planes gives

$$p_i a = d_i(a)\ell_i \quad (a \in R^2), \qquad p_i z = \left(\int_Y p_i z\right)[\mathrm{pt}] \quad (z \in R^4). \tag{41}$$

Each d_i takes the value one, and each plane evaluation takes the value one, by the same unimodular witnesses. Thus the ideal contains ℓ_0, ℓ_2 and [pt]. Its degree-four part is exactly the span of p_0, p_2 and it has no further possible degrees. This is the entire restriction kernel computed from (40), proving the ring statement. □

For every normalized index, restriction to V is consequently the class $c_k(\widehat{J}_z)$ of Proposition 4.1 reduced modulo this explicit ideal. No complement multiplication is an additional input to that formula.

5.2. **The two-plane neighborhood and its boundary.** Choose a smoothed closed regular neighborhood W of E and put $L = \partial W$, with the outward boundary orientation from the complex orientation of W. It is the plumbing of the cotangent normal disk bundles of the two planes. Their normal Euler numbers are three and their transverse intersection is positive one. The middle intersection matrix is therefore

$$B = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}. \tag{42}$$

A deleted collar retracts to L, so L models the actual overlap in the cover of M_A by V and an open enlargement of W. It is a boundary inside the resolution. The singular surface of the target passes through o; L is not the link of an isolated target singularity.

Let h_i be the positive hyperplane classes of the two components and $t_i = h_i^2$. The neighborhood ring is the wedge ring

$$h_0 h_2 = 0, \qquad h_i^2 = t_i, \tag{43}$$

with all products of degree greater than four zero. Restriction from R is

$$a \longmapsto d_0(a)h_0 + d_2(a)h_2, \qquad z \longmapsto \left(\int_Y z p_0\right) t_0 + \left(\int_Y z p_2\right) t_2 \tag{44}$$

in degrees two and four. It preserves the unit and is zero in higher positive degrees.

The Poincaré–Lefschetz sequence for (W, L) gives

$$H^k(L,\mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, 7, \\ \mathbb{Z}^2, & k = 2, 5, \\ \mathbb{Z}/8, & k = 4, \\ 0, & \text{otherwise.} \end{cases} \tag{45}$$

Indeed its only nontrivial even map in the middle is B, whose determinant is eight and whose entries have gcd one. The quotient map $\mathbb{Z}^2/B\mathbb{Z}^2 \to \mathbb{Z}/8$ is $(n_0, n_2) \mapsto n_0 - 3n_2$. Write x_i for the images of h_i and τ for the image of t_0. If ω is the oriented degree-seven generator, choose $\eta_0, \eta_2 \in H^5(L,\mathbb{Z})$ dual to x_0, x_2. The complete nonzero positive products are

$$x_0^2 = \tau, \quad x_2^2 = -3\tau, \quad x_i\eta_j = \delta_{ij}\omega, \qquad 8\tau = 0, \tag{46}$$

and $x_0x_2 = 0$. Degree and (45) make all other positive products zero. The free pairings follow from oriented Poincaré duality; the torsion products follow by restricting (43). These statements describe the ordinary ring, without asserting an additional torsion linking form.

Proposition 5.2. *With the preceding choices, restriction $\rho : H^*(V) \to H^*(L)$ is*

$$\rho^2(a) = d_0(a)x_0 + d_2(a)x_2, \qquad \rho^4([z]) = \kappa([z])\tau, \quad \kappa([z]) = \int_Y z(p_0 - 3p_2) \pmod 8. \tag{47}$$

It preserves the unit, and its other positive-degree maps vanish. For the ordered cover (V, W) with alternating map $r_V - r_W$, the Mayer–Vietoris connecting homomorphisms are

$$\partial_{\mathrm{MV}}\eta_i = -\ell_i, \qquad \partial_{\mathrm{MV}}\omega = -[\mathrm{pt}]. \tag{48}$$

Proof. Restrictions from M_A to the overlap agree whether taken through V or W. Surjectivity in Proposition 5.1 and (44) therefore give (47). It is well-defined: evaluation of $p_0 - 3p_2$ against p_0, p_2 gives $0, -8$. Multiplicativity may also be checked without referring back to the geometric map. For $a, b \in R^2$,

$$\int_Y abp_i = d_i(a)d_i(b),$$

which is exactly the product prescribed by (46). All further positive products have zero target. This checks the map on the complete integral basis and every ring relation.

For signs, the pair boundary of (W, L) sends the degree-five classes dual to h_i to the positive relative line classes, and its oriented top class to the positive relative point class, by Stokes' convention with outward boundary. Excision identifies their images in M_A with $\ell_i, [\mathrm{pt}]$. In the ordered alternating restriction $r_V - r_W$, a cochain supported on the W side has the negative of that pair boundary. This yields (48). □

In degree four the square recovers the additive compatibility description

$$R^4 \simeq \left\{ ([z], n_0, n_2) : \ n_0 - 3n_2 \equiv \int_Y z(p_0 - 3p_2) \pmod 8 \right\}, \tag{49}$$

where $[z] \in (R/(p_0, p_2))^4$. The ambient ring was already constructed in Section 4; (49) is not a separate derivation of multiplication from an additive sequence.

For the normalized indices, (17) gives

$$\begin{aligned} \kappa(c_2(I_z^y)) &= \binom{m_0(z)}{2} - 3\binom{m_2(z)}{2} \pmod 8, \\ \kappa(c_1(I_z^y)c_1(I_{z'}^y)) &= m_0(z)m_0(z') - 3m_2(z)m_2(z') \pmod 8. \end{aligned} \tag{50}$$

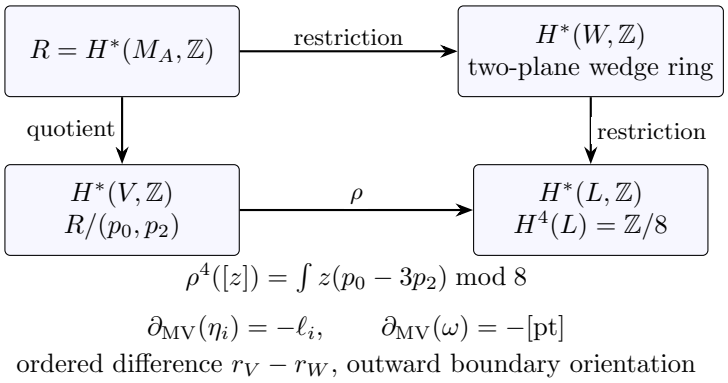


FIGURE 3. The computed restriction square in chamber A. The lower map in degree four is κ and the right map is the cokernel of B. The signed connecting maps belong to the ordered Mayer–Vietoris sequence and are separate from these ring homomorphisms.

These reproduce the earlier binomial attachment of [Wue26a, §6.4]. The present formula also calculates its full domain: evaluate on A, T_i, C_{ij}, Q_i using (26), with p_i expanded by (27). This is a specified integral generating set with all its products, not a test on a possibly proper sublattice of degree four.

5.3. **A uniform formula for all six chambers.** In chamber D the two planes have the same surface classes and the negative line classes. Thus use the preceding formulas with degree-two restriction $a \mapsto \epsilon d_0(a)h_0 + \epsilon d_2(a)h_2$, where $\epsilon = 1$ in A and -1 in D. The relative maps in degrees $4, 6, 8$ are

$$(p_0, p_2), \qquad (\epsilon\ell_0, \epsilon\ell_2), \qquad [\mathrm{pt}]. \tag{51}$$

The boundary ring and degree-four functional are unchanged, and the connecting maps are the negatives of the displayed line and point maps.

For the four ruled configurations use

σ	B	C	E	F
(k, j, ϵ)	$(2, 0, -1)$	$(0, 2, 1)$	$(2, 0, 1)$	$(0, 2, -1)$.

(52)

The surfaces are p_k and $p_j - p_k$. Write e_Σ for the negative section and r_Σ for the ruling in $\Sigma \simeq \mathbb{F}_1$. Their global curve classes are

$$[e_\Sigma] = \epsilon c_k, \qquad [r_\Sigma] = -\epsilon(c_0 + c_2), \qquad e_\Sigma^2 = -1, \quad e_\Sigma r_\Sigma = 1, \quad r_\Sigma^2 = 0. \tag{53}$$

Let $\xi, \upsilon \in H^2(W_\sigma)$ take values $(1, 0)$ and $(0, 1)$ on (e_Σ, r_Σ). Their restrictions to the plane are $h, 0$ and to Σ are $r_\Sigma, e_\Sigma + r_\Sigma$, respectively. With t_P, t_Σ dual to the two surface components, the ring is

$$\xi^2 = t_P, \qquad \xi\upsilon = \upsilon^2 = t_\Sigma, \tag{54}$$

with higher positive products zero. This follows from the Mayer–Vietoris ring of the two components meeting along their common curve, or directly from the two restrictions.

For $a \in R^2$ set $s = \epsilon d_k(a)$ and $t = -\epsilon(d_0(a) + d_2(a))$. The complete positive restrictions to the neighborhood are

$$\begin{aligned} a|_{W_\sigma} &= s\xi + t\upsilon, & a|_\Sigma &= te_\Sigma + (s+t)r_\Sigma, \\ z|_{W_\sigma} &= \left(\int_Y zp_k\right) t_P + \left(\int_Y z(p_j - p_k)\right) t_\Sigma. \end{aligned} \tag{55}$$

The relative middle matrix is $B' = \left(\begin{smallmatrix} 3 & -2 \\ -2 & 4 \end{smallmatrix}\right)$ from (22). Its cokernel map is $(n_P, n_\Sigma) \mapsto 6n_P + n_\Sigma \pmod 8$. The boundary has the groups (45). Write τ_k for the image of t_Σ and retain the symbols ξ, υ for their boundary images. Its nonzero positive products are

$$\xi^2 = 6\tau_k, \quad \xi\upsilon = \upsilon^2 = \tau_k, \quad \xi\eta_\xi = \upsilon\eta_\upsilon = \omega, \quad 8\tau_k = 0, \tag{56}$$

where η_ξ, η_υ are the dual degree-five basis. Their cross pairings are zero. Every other positive product vanishes.

Proposition 5.3. *In every ruled row of (52), restriction from the complement is*

$$\rho^2(a) = s\xi + t\upsilon, \qquad \rho^4([z]) = \left(\int_Y z(p_j + 5p_k)\right)\tau_k \pmod 8. \tag{57}$$

Its relative maps in degrees $4, 6, 8$ *are*

$$(p_k, p_j - p_k), \qquad (\epsilon\ell_k, -\epsilon(\ell_0 + \ell_2)), \qquad [\mathrm{pt}]. \tag{58}$$

For the ordered difference $r_V - r_W$, the connecting maps on $\eta_\xi, \eta_\upsilon, \omega$ are the negatives of the corresponding line and point maps. All six complement rings are $R/(p_0, p_2)$.

Proof. Proposition 3.1 and (21) give these as the actual adjacent compact configurations. The two surface classes generate the same ideal (p_0, p_2), and their line pair is a unimodular change of ℓ_0, ℓ_2. The relative groups and primitive maps therefore give the same quotient proof as in Proposition 5.1. Their regular neighborhoods have the computed component ring and intersection matrix, so Poincaré–Lefschetz duality gives (56) and (57). The actual compact maps identify the exteriors, and regular levels outside the exceptional core identify their outward boundaries. Thus these are the overlap maps of the given geometric gluing.

To check integrality and multiplicativity explicitly, evaluation of $p_j + 5p_k$ against p_j, p_k gives $8, 16$, so the degree-four map kills the ideal. For $a, b \in R^2$,

$$\int_Y abp_k = d_k(a)d_k(b), \qquad \int_Y ab(p_j - p_k) = d_j(a)d_j(b) - d_k(a)d_k(b).$$

Substitution of s, t in (56) gives $6s_as_b + s_at_b + t_as_b + t_at_b$, which is the same value modulo eight. All other positive products have zero target. The relative maps are the geometric component, line and point inclusions. Their connecting signs follow from the same outward Stokes convention as (48). □

For $k = 2$ the functional in (57) is κ, while for $k = 0$ it is 5κ. This change records the torsion-generator convention, not a new uncomputed gluing choice. For example across $A \to B$, the degree-two boundary classes transform as

$$x_0 \longmapsto \upsilon, \qquad x_2 \longmapsto \upsilon - \xi.$$

The degree-five classes transform by the inverse transpose: $\eta_0 \mapsto \eta_\xi + \eta_\upsilon$, $\eta_2 \mapsto -\eta_\xi$, and $\omega \mapsto \omega$. These formulas preserve (46) and (56), with $\tau \mapsto \tau_2$.

5.4. **The moving divisor remains in the complement.** The open set V_σ contains the divisor D° over S°, so it is larger than $\pi_\sigma^{-1}(U)$. We record its established ring to make this distinction explicit. In chamber A use x from (14), set $z_D = a_1 + 2x$, and put

$$d = \theta_w(z_D)|_{S^\circ}, \qquad h_D = \theta_v(x)|_{D^\circ}.$$

The open-divisor calculation of [Wue26a, §7] is

$$\begin{aligned} H^2(S^\circ,\mathbb{Z}) = H^2(S,\mathbb{Z})/\mathbb{Z}[C] \simeq \mathbb{Z}^{21}, \qquad H^k(S^\circ,\mathbb{Z}) = 0 \quad (k>0,\ k\neq 2),\\ H^*(D^\circ,\mathbb{Z}) = H^*(S^\circ,\mathbb{Z})[h_D]/(h_D^2 - dh_D), \qquad D|_{D^\circ} = -2h_D + \pi^* d. \end{aligned} \tag{59}$$

Here $[C]$ is primitive, so the Gysin sequence gives the base cohomology. The PGL_2 structure group acts trivially on fibre cohomology, and h_D has fibre degree one, giving an integral Leray–Hirsch basis. The topological rank-two lift exists because $H^3(S^\circ,\mathbb{Z}) = 0$; normalizing its hyperplane class to h_D gives first Chern class d and zero second Chern class. The universal extension gives the divisor identity; the orthogonality of z_D to a_1, w, v removes base-line ambiguities.

The algebraic Brauer obstruction remains nonzero of order two on S°: restriction to the function field is injective for a regular integral surface. Its analytic image on this open surface vanishes, using $H^2(S^\circ,\mathcal{O}) = H^3(S^\circ,\mathbb{Z}) = 0$. These lifting statements and their primary sources are proved in [Wue26a, §7.1]. They do not imply an algebraic universal family over the full compact S. The divisorial wall proof used coarse blow-ups precisely so that such a lift was unnecessary.

6. The wall actions in integral coordinates

The compact geometric actions were proved in Proposition 3.3. We now give their coordinates on every degree of R and on the normalized indices. This also verifies compatibility with the attachment formulas.

6.1. **All ring generators.** At a divisorial wall the action in R is the conjugate of s_{a_1} by g. Put

$$k_0 = 2H - D_2, \qquad r = \mu(k_0) + \delta = \theta_{v'}(ga_1), \qquad q(k_0) = 0, \quad q(r) = -2. \tag{60}$$

Write S for this integral ring automorphism. Its degree-two action is

$$\mathsf{S}(\mu(b)) = \mu(b) + q(b,k_0)r, \qquad \mathsf{S}(\delta) = \delta - 2r. \tag{61}$$

Set $\mu_i' = \mathsf{S}(\mu_i)$, $\delta' = \mathsf{S}(\delta)$. The full degree-four action is

$$\begin{aligned} A' &= \frac{c_2(Y) + 3(\delta')^2}{24} = A + \frac{\mu(k_0)^2 + \delta\mu(k_0)}{2},\\ T_i' &= \mu_i'\delta', \qquad C_{ij}' = \mu_i'\mu_j' - q_{ij}A',\\ Q_i' &= \frac{(\mu_i')^2 - q_{ii}A' - T_i'}{2}. \end{aligned} \tag{62}$$

In degree six the images are $\mathsf{S}(R_i) = \mathsf{S}(\mu(\alpha^i))A'$ and $\mathsf{S}(R_\delta) = -\delta' A'$. The unit and oriented point are fixed. Formula (62) follows from preservation of $c_2(Y)$ and (27). Its apparently rational expressions are integral because they are images under the integral ring map already constructed; the second formula for A' also shows its integrality directly. These identities give a finite coordinate calculation on the entire lattice, including the classes outside its integral symmetric square.

They imply

$$\mathsf{S}(p_0) = p_2, \quad \mathsf{S}(p_2) = p_0, \quad \mathsf{S}(c_2(Y)) = c_2(Y), \quad \mathsf{S}^2 = \mathrm{id}\,. \tag{63}$$

Consequently S preserves the exceptional ideal. In the fixed two-plane torsion convention,

$$\kappa(\mathsf{S}z) = 5\kappa(z) \pmod 8,$$

because $p_2 - 3p_0 \equiv 5(p_0 - 3p_2)$ modulo eight in the integral lattice. The ruled conventions in (57) incorporate exactly this change. At a plane wall every generator has the identity action in R, with the geometric surface change (21) still understood. It is the exceptional product in (20) that makes this the action of a multiplicative correspondence.

6.2. **The fixed determinant normalizer.** Let $h = \mathrm{id}$ or s_{a_1} act between the source and target Mukai lattices, and retain the original y on both sides. Markman's universal-class identity, followed by the rank-one normalization argument, gives

$$\gamma_h\big(c_k(I^y_z)\big) = c_k\left(I^y_{hz} \otimes (\det I^y_{hy})^{-1}\right). \tag{64}$$

Indeed transport first gives the target universal family normalized by hy rather than by y. Starting from the y-normalized target family, its hy-index still has rank one, so tensoring by its inverse determinant is exactly that change of normalization. Equality of characters implies equality of integral Chern classes by torsion-freeness, as in Proposition 4.1.

For $h = \mathrm{id}$ the determinant is trivial. For $h = s_{a_1}$, $(y, a_1) = -1$ and hence $hy = y - a_1$. Additivity of the index and $\det I^y_y = 1$ give

$$c_1\left((\det I^y_{y-a_1})^{-1}\right) = \theta_v(a_1), \qquad \Phi\theta_v(a_1) = r. \tag{65}$$

Thus in the common ring the transformed character is $\mathrm{ch}(I^y_{s_{a_1}z})\exp(r)$, not merely the character with its input vector reflected. Formula (36) and the Newton identities calculate all its Chern classes. Functoriality includes these normalizing lines, so they introduce no ambiguity in the six-step cohomological identity.

6.3. **Dependence on the choices.** An integral change of basis of Λ changes the coordinates in (25) and (28) by the corresponding integral transformation. The diagonal construction itself is intrinsic. The isometry g and the universal normalizer y are specified: Markman's functoriality fixes the transport, and (64) describes any change of the normalizer. Different geometric realizations of an action with the same degree-two marking have the same cohomological action, by rational generation and torsion-freeness.

Isotopic regular neighborhoods give the same restriction and relative maps by naturality of excision. The outward orientation and the ordered difference $r_V - r_W$ fix the connecting signs. Replacing a torsion generator by a unit multiple changes the printed mod-eight functional by that unit. These choices specify the calculated cohomology and its maps; no assertion about a homotopy classification of the gluing follows from them.

Proof of Theorem 1.1. Proposition 3.1 constructs the compact partial-wall maps, and Lemma 3.2 and Proposition 3.3 identify their integral actions. Equations (32)–(38) give the specified ambient marking and exceptional coordinates. Propositions 5.1, 5.2 and 5.3 calculate the complement and every attachment map. Proposition 4.1 and (64) give the universal-index markings and their determinant correction. Finally Corollary 3.4 proves the cohomological path relation. All computations use the same compact moduli problem and the same primitive cycles. □

6.4. **Exact verification.** The formulas above are finite expressions in the marked K3 Gram matrix. Accompanying standard-library Python programs implement (23) with rational arithmetic. They compute the integral middle pairing, the two plane coordinates, the four Chern classes of each index for a 24-element topological K-basis, and the images of every degree-two, degree-four and degree-six basis element under S. Additional virtual classes test negative ranks and the determinant normalization. The retained outputs check the plane binomial restrictions, all six degree-two attachment products, and the complete wall loop. The printed formulas define the calculation independently of these programs; the compact wall and topological-kernel arguments are the geometric justifications for using them.

7. The calculated cycles in harmonic geometry

The wall correspondences identify integral classes. Their metric interpretation uses the harmonic metrics of the same compact bundle moduli, with the normalization in [Wue26a, Wue26b]. We first calculate exact periods, then state precisely which existing degeneration results apply.

7.1. **Periods, volumes and specialization.** Let ω_η be the smoothly varying Ricci-flat representative of $H+sD_1+tD_2$. Fix the holomorphic determinant and use its smoothly varying Hermitian–Einstein connections, together with the real Dolbeault harmonic L^2 metric, unnormalized by the volume of X. Its Kähler class is

$$[\Omega_\eta] = 2\pi^2\theta_v(b_\eta), \qquad b_\eta = (0, [\omega_\eta], c_1(v)\cdot[\omega_\eta]/3), \tag{66}$$

by [Wue26a, §2.4]. Hence the period of a curve functional c is $2\pi^2(c, b_\eta)$ and a surface volume is half its pairing with $[\Omega_\eta]^2$.

In chamber A put $a=\zeta_0>0$, $b=\zeta_2>0$. Equations (4), (12) and (66) give

$$\begin{gathered}\operatorname{Area}(c_0)=\pi a, \qquad \operatorname{Area}(c_2)=\pi b, \qquad \operatorname{Area}(f)=\pi(a+b),\\ \operatorname{Vol}(P_0)=\tfrac{1}{2}\pi^2a^2, \qquad \operatorname{Vol}(P_2)=\tfrac{1}{2}\pi^2b^2.\end{gathered} \tag{67}$$

The equality $f=c_0+c_2$ is the actual proper reduced specialization from [Wue26a, §5.1]. It is preserved, with the signs of Table 1, by the compact wall maps. In a ruled row the plane line and common section have area $\pi|\zeta_k|$, and the ruling has area $\pi|\zeta_1|$. The class p_j-p_k therefore has volume

$$\operatorname{Vol}(\Sigma) = \begin{cases} \tfrac{1}{2}\pi^2(\zeta_0^2-\zeta_2^2), & \sigma = B, E,\\ \tfrac{1}{2}\pi^2(\zeta_2^2-\zeta_0^2), & \sigma = C, F.\end{cases} \tag{68}$$

Both expressions are positive in their sectors. Alternatively, restrict $[\Omega_\eta]$ using (55) and compute its square on $\mathbb{F}_1$; this gives the same values.

For a wall action $h=\mathrm{id}$ or s_{a_1} the exact period identity is

$$2\pi^2(hc, b_{\eta'}) = 2\pi^2(c, h^{-1}b_{\eta'}). \tag{69}$$

This is a pairing identity, not an assertion that the wall map is a metric isometry. In particular $h^{-1}b_{\eta'}$ need not be a polarization covector in the chosen two-parameter family.

7.2. **Hypotheses of the metric comparison.** The analytic results used here have the following hypotheses, verified for this family in [Wue26b, §9.1]. They are the conditions (G1)–(G3) of that paper:

(1) The finitely many smooth projective chamber spaces equal their generic Gieseker moduli and have proper surjective maps to the connected normal projective central target. There is a connected dense stable open, a finite Whitney refinement of factor types, and agreement of the projective and fixed-wall gauge topologies. The maps are isomorphisms over the stable open. Every chamber object is H-semistable; the central map assigns its H-graded class, and its complex morphism is constant within a chamber.
(2) Every central semistable sheaf is a bundle and every central ideal HYM object has zero lost charge. Varying-metric surface compactness gives smooth gauge subsequences and excludes nearby charge.
(3) The full convergent doubled Ext-quiver charts retain all tangent factors, loops and multiplicities. Each complex moment-zero fibre contains a module-simple point, meaning a simple representation of its quiver with relations. The classifying germs commute with the projective-to-affine quotient and the central graded-class maps.

The norm on every Ext space is its actual central harmonic norm. At a central bundle $E_b = \bigoplus_i F_i \otimes \mathbb{C}^{m_i}$, let $W_b = \operatorname{Ext}^1(E_b, E_b)$ be the full harmonic representation and $K_b = (\prod_i \mathrm{U}(m_i))/\mathrm{U}(1)$ its effective compact stabilizer. Denote the full quotient and its intrinsic stratified length metric by

$$\mathcal{Q}_{b,\nu} = \{\mu_{\mathbb{C}} = 0,\ \mu_{\mathbb{R}} = \nu\}/K_b, \qquad d_{b,\nu}.$$

Paths may be finitely piecewise smooth across representation-type strata. Its radius r_ν is the norm of a balanced representative. The moment levels use ordinary multiplicity trace:

$$\zeta_{b,i}(\eta) = 2\pi\big(r_i\mu_\eta(v) - \deg_\eta F_i\big), \qquad r_i = \operatorname{rank} F_i, \qquad \sum_i m_i\zeta_{b,i} = 0. \tag{70}$$

For $L_1 \oplus F$ the transverse two-arrow level is $2\pi(2\mu_\eta(v) - \deg_\eta F) = -\zeta_1$. Ordinary multiplicity trace is distinct from the rank-weighted mass in small-spectrum statements.

For each fixed background the actual classifying chart $\Phi_{b,\eta}$ satisfies the squared-metric estimate

$$(1-\varepsilon_b)g_{b,\zeta_b(\eta)} \le \Phi_{b,\eta}^* g_\eta \le (1+\varepsilon_b)g_{b,\zeta_b(\eta)}, \qquad \varepsilon_b = C_b(|\eta| + \|x\|^2). \tag{71}$$

This is [Wue26a, Corollary 2.2], with whole-preimage coverage as in [Wue26b, Proposition 5.1]. The constant depends on the fixed background and complementary elliptic gap, but not on ratios of positive finite orbit eigenvalues. A fixed deeper chart handles degeneration of shallower backgrounds. No uniform gap is assumed for separate backgrounds approaching a reducible bundle.

7.3. **Ambient collapse and complete rescalings.** The diameter statements in [Wue26a, §8] use ambient distances in the full compact moduli, not only the intrinsic metrics of the exceptional surfaces. If $\rho = a + b$ in chamber A, they give

$$\operatorname{diam}_{g_\eta} P_i = \sqrt{\rho}\,(1 + O(\rho))D_i(a/\rho, b/\rho), \tag{72}$$

where D_i is the ambient diameter of the corresponding plane in the complete normalized quotient. The finite quotient diameters obey, uniformly in the positive ratio,

$$\sqrt{2a} \le \operatorname{diam}_{\mathrm{model}} P_0 \le \frac{\pi}{\sqrt{2}}\sqrt{a},$$

and the same inequalities hold with b for P_2. The actual diameter has relative factor $1+O(\rho)$. As $\eta \to 0$, if $a/b \to 0$, then $\operatorname{diam} P_0/\sqrt{a} \to \pi/2$; the reciprocal statement holds for P_2. For

moving $\mathbb{P}^1$ fibres over any compact subset of S°, $\operatorname{diam}/\sqrt{|\zeta_1|} \to \pi/2$ uniformly as $\eta \to 0$ through nonzero transverse levels ζ_1.

The passage to ambient distances is an analytic input. With squared-metric bounds $1/2, 2$, a path starting at radius $r_0 < R$ costs at least $(R - r_0)/\sqrt{2}$ before exiting radius R. This first-exit estimate confines short competitors to the appropriate chart; comparison of both length infima then yields the relative errors even for a much smaller plane. These arguments apply to our computed classes because their cycle realization and classifying charts are the same ones.

We state the complete rescaling hypotheses to specify the meaning of this compatibility. Fix a central chart and suppress b. Suppose $\eta_n \to 0$ is generic, $\delta_n \downarrow 0$, and

$$\nu_n := \zeta_b(\eta_n)/\delta_n^2 \longrightarrow \nu. \tag{73}$$

Choose balanced representatives p_n at these normalized levels which converge to p after compact group adjustment. Then [Wue26a, §8.3 and Theorem 8.1] gives

$$(M_{\sigma(\eta_n)}, \delta_n^{-1} d_{g_{\eta_n}}, \Phi_{b,\eta_n}([\delta_n p_n])) \longrightarrow (\mathcal{Q}_{b,\nu}, d_{b,\nu}, [p]) \tag{74}$$

in complete pointed Gromov–Hausdorff topology. On bounded rescaled balls the relative error against the current finite quotient is $O_L(|\eta_n| + \delta_n^2)$. This is not a rate for $\mathcal{Q}_{b,\nu_n} \to \mathcal{Q}_{b,\nu}$.

For a further scale, let x_* represent the limiting point at level $\nu_* = \lim \nu_n$, with stabilizer K. Decompose its full hyperkähler normal representation as $W = T \oplus W'$, where $T = W^K$ and $W' = T^\perp$ for the retained invariant inner product. In this slice write the current centers as $(u_n, z_n) \to 0$ and require

$$\sigma_n \downarrow 0, \qquad \frac{(\nu_n - \nu_*)|_{\operatorname{Lie} K}}{\sigma_n^2} \longrightarrow \lambda, \qquad z_n/\sigma_n \longrightarrow z_*. \tag{75}$$

At physical scale $\delta_n \sigma_n$ the complete limit is $T \times (W' /\!\!/_\lambda K)$, based at $(0, [z_*])$. Tangential translation removes u_n; no bound on $|u_n|/\sigma_n$ is required. The additional slice error on a fixed bounded ball is $O_{x_*}(|\nu_n - \nu_*| + |u_n| + \sigma_n R)$. Inverse coverage and the outer and inner exit costs give whole actual balls at each stage. This is the finite-stage theorem of [Wue26a, Theorem 8.1] and [Wue26b, Proposition 8.3]; any fixed finite iteration must retain the same normalized-level and normal-basepoint data at every stage.

At a nonzero ζ_0 or ζ_2 wall the normal resolution is $T^*\mathbb{P}^2$. At a nonzero ζ_1 wall the full model has a harmonic $\mathbb{C}^2$ tangent factor and the A_1 transverse quotient, resolved by the Eguchi–Hanson metric at nonzero level. An $\mathbb{F}_1$ component belongs to the interacting quotient; it is not asserted to have a complete cotangent-$\mathbb{F}_1$ limit. Table 1 identifies the effective cycles in each of these inherited models.

7.4. **Global convergence with the same contractions.** Under the three stated hypotheses, (Z, d_0) is the compact completion of the central harmonic metric on U. We recall the quantitative form of [Wue26b, Theorem 5.5], since it applies to the same π_η used in our integral calculation. Choose $\varepsilon_j \downarrow 0$, $\varepsilon_j < 1/8$. For each accuracy choose a finite central chart cover with outer radii $16 r_{ji}$, inner radii $r_{ji}/4$, and squared-metric error at most ε_j. Then shrink a parameter neighborhood $\mathcal{P}_j$ so that whole-preimage coverage holds.

For a finite quotient contraction $q_\zeta : \mathcal{Q}_\zeta \to \mathcal{Q}_0$ retain the distance discrepancy

$$b_{i,r}(\eta) = \sup_{\substack{r_0(z), r_0(z') \le r \\ q_\zeta(x) = z,\ q_\zeta(x') = z'}} |d_\zeta(x, x') - d_0(z, z')|$$

and both radial discrepancies

$$a_{i,r}(\eta) = \sup_{\substack{q_\zeta(x)=z \\ r_0(z)\le r}} |r_\zeta(x) - r_0(z)|,$$
$$\widetilde{a}_{i,r}(\eta) = \sup_{r_\zeta(x)\le r} |r_0(q_\zeta(x)) - r_\zeta(x)|.$$

The symbols d_0, r_0 in these formulas are the finite zero-quotient distance and radius. Properness and intrinsic parameter continuity make all three finite and tending to zero. Arrange $a, \widetilde{a} < r/16$ and $b < r$. Choose

$$0 < \ell_j < \min_i \frac{r_{ji}}{16\sqrt{2}}, \quad \alpha_j = \sqrt{\frac{1+\varepsilon_j}{1-\varepsilon_j}} - 1, \quad \beta_j(\eta) = \sqrt{1+\varepsilon_j}\, \max_i b_{i,r_{ji}}(\eta).$$

A fixed finite central cover supplies $D_* \ge \operatorname{diam}(Z, d_0)$. The stated theorem then gives

$$\operatorname{dis}(\operatorname{graph} \pi_\eta) \le \inf_{j:\eta\in\mathcal{P}_j} \left[\left(\alpha_j + \frac{\beta_j(\eta)}{\ell_j} \right) D_* + \beta_j(\eta) \right] \longrightarrow 0. \tag{76}$$

The Gromov–Hausdorff distance is bounded by half this quantity. The proof uses the two radial discrepancies to compare short ambient pairs in both directions, and then subdivides paths into lengths at most ℓ_j. For fixed j, first let $\eta \to 0$; then let the accuracy index tend to infinity. No universal power law is asserted for this modulus.

The period calculation and the cycle table identify the integral classes entering these metric results. They leave the harmonic norms, no-charge compactness, convergent charts and normalized basepoint data as analytic hypotheses. Bubbling, unrestricted HYM dynamics and genuine complex smoothings require additional results.

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